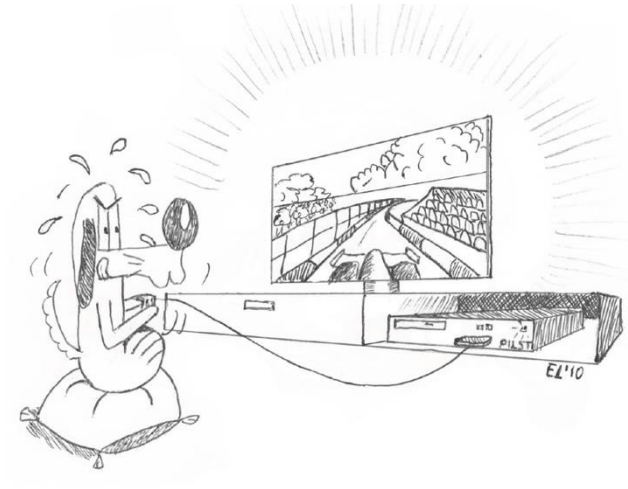
ΣΥΣΤΗΜΑΤΑ ΑΝΑΜΟΝΗΣ
(Queuing Systems)

Simulation



Book from where to read:

Performance Evaluation of Computer and Communication Systems

Slides material adapted from Performance Evaluation Course, Jean-Yves Le Boudec, EPFL

Contents

1. What is a simulation ?

2. Accuracy of simulation output

3. Pseudo Random Number Generators

4. How to sample

5. Monte Carlo

What is a simulation ?

- Experiment in the computer: write programs to immitate reality

<u>Differences between simulation/implementation:</u>

1. Events will serialize in simulation

2. Simulated vs. real time: simulated time can move much faster (e.g., when there are no arrivals in a queue) or much slower (e.g., when there are many concurrent arrivals in a queue)

3. Initialization: impact of initial system state e.g., start from a completely empty system?

Two different simulation examples

Simulation 1: Time to complete N jobs

N customers arrive (at the same time) at Joe's shop and download a file. We want to estimate the time T it takes to serve the N customers

Terminating simulation: Well-defined beginning and ending of the simulation

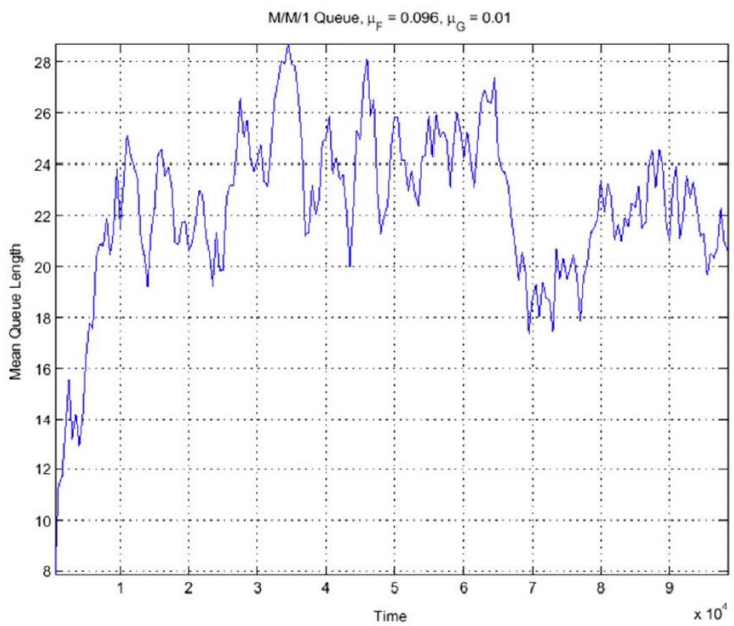
Simulation 2: Delay at a server

An information server handles requests according to some scheduling policy. We want to estimate the time it takes to serve one request.

Not terminating simulation.
Practical question: How long should we run it?

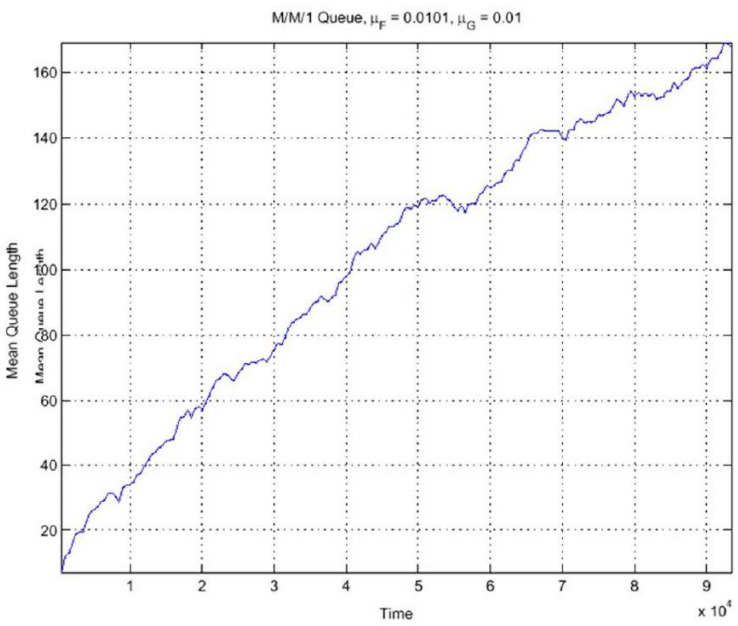
Two simulation runs
What is the difference ?

Scenario 1



Average service time = 96 ms

Scenario 2



Average service time = 101 ms

Request arrival rate = 10 /s

Deficit of 1% between service and arrival time

Definition of stationarity – Reminder

! Important issue for non-terminating simulations

✓ A property of a stochastic process X_t

Let X_t be a stochastic process that represents the state of the simulator at time t.

It is stationary iff:

for any $s > 0$
$(X_{t_1}, X_{t_2}, X_{t_3}, \dots, X_{t_n})$ has the same distribution as $(X_{t_1+s}, X_{t_2+s}, X_{t_3+s}, \dots, X_{t_n+s})$

i.e., there is no way to differentiate between samples of these two tuples

i.e., the simulation does not get old.

Classic cases

Markov Models

The state X_t is sufficient to draw the future of the simulation

For a Markov model over a discrete state space (quantified by integers), such as our queue models:

If you run the simulation long enough it will either walk to infinity (unstable) or converge to a stationary regime

Example:

An M/M/1 queue with $\rho > 1 \quad unstable$

An M/M/1 queue with $\rho < 1$ → becomes stationary after transient

If the state space is strongly connected (any state can be reached from any state) then there is 0 or 1 stationary regime

Example: queue

Else, when there exist irreversible transitions, there may be several distinct stationary regimes

Example: system with failure modes when you do not model repair

Stationarity and Transience

- Knowing whether a model has a stationary regime is sometimes a hard problem

- We will see important models where this is solved theoretically
  Ex: some queuing systems, e.g., M/M/1 queue where utilisation factor lower than 1 means stationarity
  Ex: time series models

- Reasoning about your system may give you indications
  Do you expect growth ?
  Do you expect seasonality ?

- Once you believe your model is stationary, you should handle transients in order to eliminate the impact of the initial conditions
  Remove (how? Look at your output and guess)
  Sometimes it is possible to avoid transients at all (perfect simulation)

Typical reasons for non-stationarity

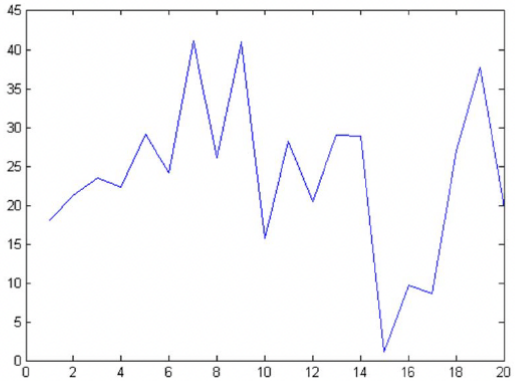
- Obvious dependency on time
  Seasonality, growth
  Can be ignored at small time scale (minute)

- Non-Stability: Explosion
  Resources, e.g., M/M/1 queues with utilization factor >1

- Non-Stability: Freezing Simulation
  System becomes slower with time (aging)
  Typically, because there are rare events of large impact («Kings»), e.g., long jobs
  The longer the simulation, the larger the largest king

Question 1

The state of a simulation at time $\boldsymbol{t = 1, 2, 3, \ldots}$ is drawn at random from a distribution F(). Is this a stationary simulation ?



X_t is a sample drawn from the distribution $N(23, 100)$

A. Yes

B. It depends on the distribution $F(\quad)$

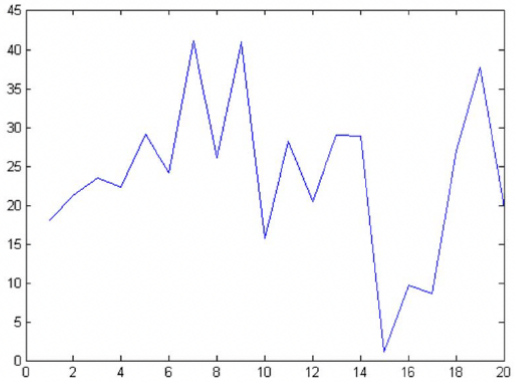
C. It depends if the simulation terminates or not

D. No

E. I don't know

Question 1 - Solution

The state of a simulation at time $\boldsymbol{t = 1, 2, 3, \ldots}$ is drawn at random from a distribution F(). Is this a stationary simulation ?



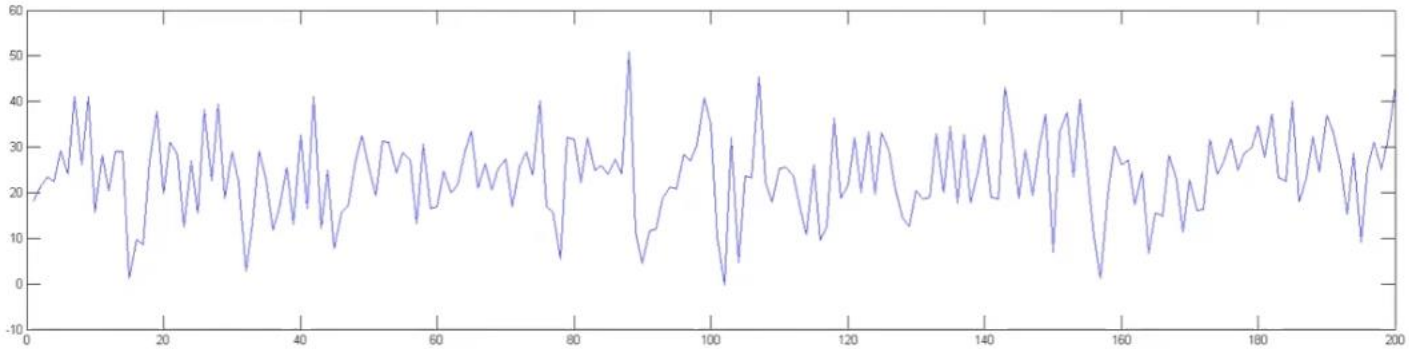
X_t is a sample drawn from the distribution $N(23, 100)$

A. Yes
B. It depends on the distribution $F(\quad)$
C. It depends if the simulation terminates or not
D. No
E. I don't know

Question 1 – Solution

- ✓ It is statistically impossible to see whether a part of the simulation is young or old.

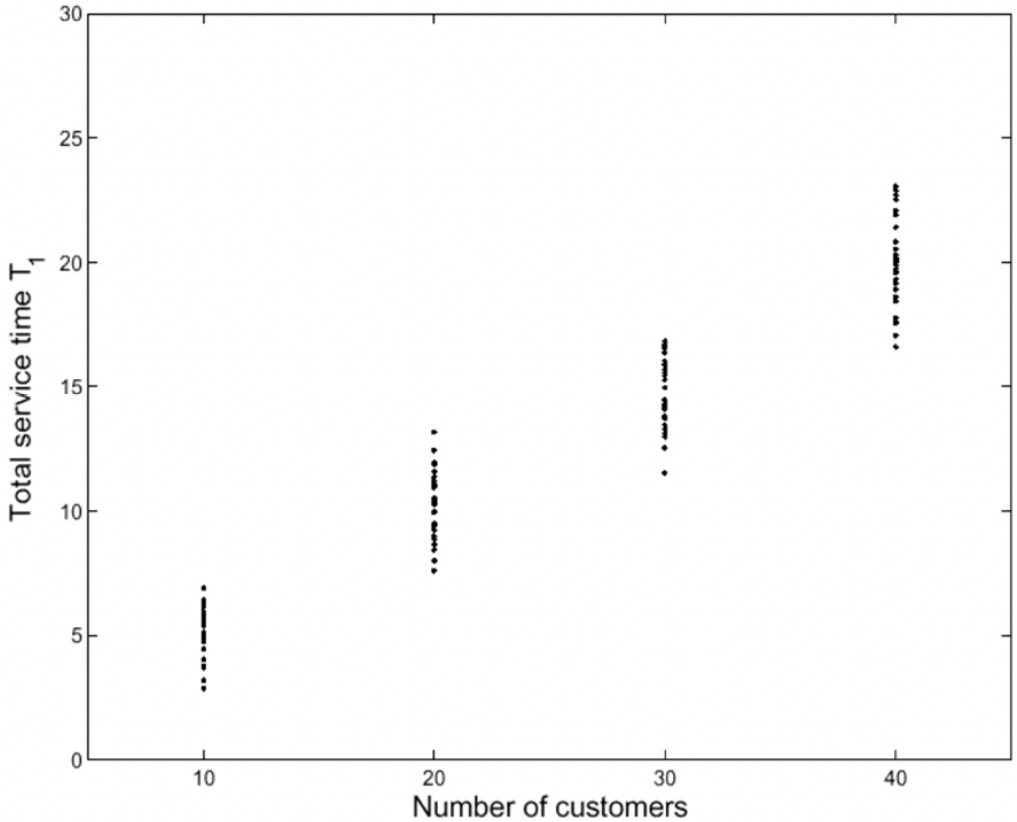
- ✓ The joint distribution of $(X_{t_1}, X_{t_2}, X_{t_3}, \dots, X_{t_n})$ is the same distribution as $(X_{t_1+s}, X_{t_2+s}, X_{t_3+s}, \dots, X_{t_n+s})$ for all values of s. *This is the distribution of a vector of n independent random variables drawn independently from F().*



Accuracy of Simulation Output

A stochastic simulation produces a random output → we need confidence intervals
Uses replication

30 runs of Joe's shop: estimate the time to serve a batch of N customers



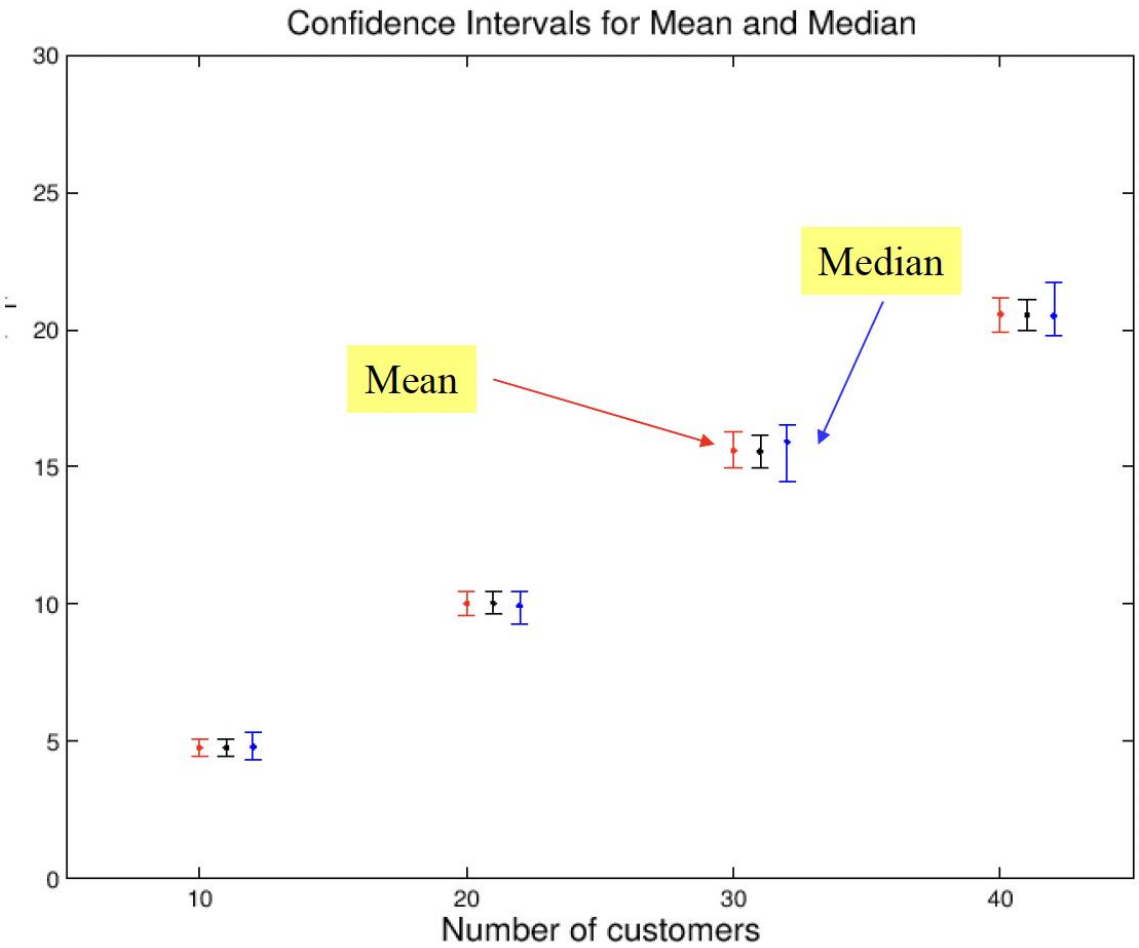
Terminating simulation so no issue with stationarity. However is random.

Confidence Intervals

Number + quantification of uncertainty of that number for previous slide's simulation

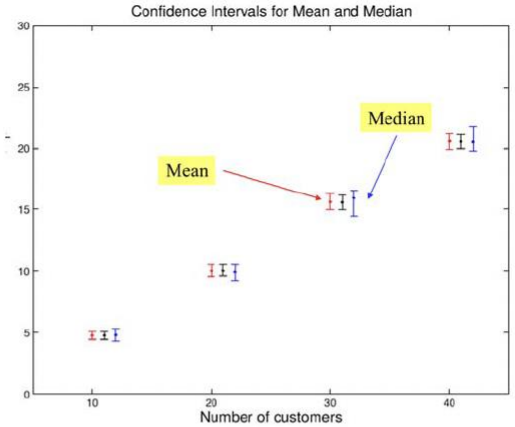
Red: mean + confidence interval for mean

Blue: median+ confidence interval for median



Question 2

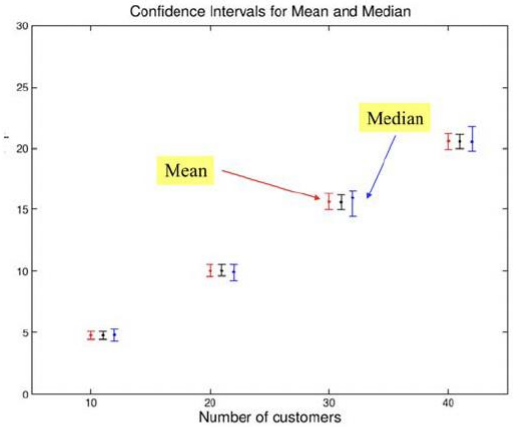
When computing these confidence intervals we need to make sure that…



A. … the simulation runs are independent
B. … the output is normally distributed
C. Both
D. None
E. I don't know

Question 2

When computing these confidence intervals we need to make sure that…



A. … the simulation runs are independent

B. … the output is normally distributed → *Desirable but asks for too much*

C. Both

D. None

E. I don't know

Pseudo Random Number Generator

Key to produce independent replications of a simulation run

A stochastic simulation does not use truly random numbers but pseudo random numbers.

The Pseudo Random Number Generator produces a random number ~ *U(0,1)* based on a chaotic sequence: sequence of numbers that appear to be random although it is deterministic.

EXAMPLE 6.8: LINEAR CONGRUENCE. A widespread generator (for example the default in ns2) has $a = 16'807$ and $m = 2^{31} - 1$. The sequence is $x_n = \frac{sa^n \bmod m}{m}$ where s is the seed. m is a prime number, and the smallest exponent h such that $a^h = 1 \bmod m$ is $m - 1$. It follows that for any value of the seed s, the period of x_n is exactly $m - 1$.Figure 6.5 shows that the sequence x_n indeed looks random.

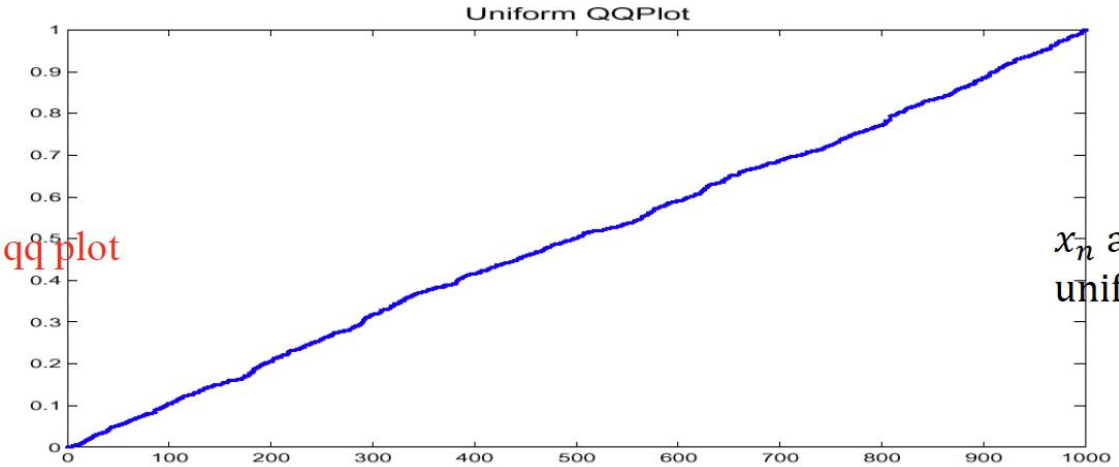
Most often used formula $X_{n+1} = aX_n mod\ m, X_0 = s$

Let us check if output looks random (next slide)

The Linear Congruential Generator of ns2

X axis: quantiles of uniform distribution
Y axis: quantiles of the sampled sequence

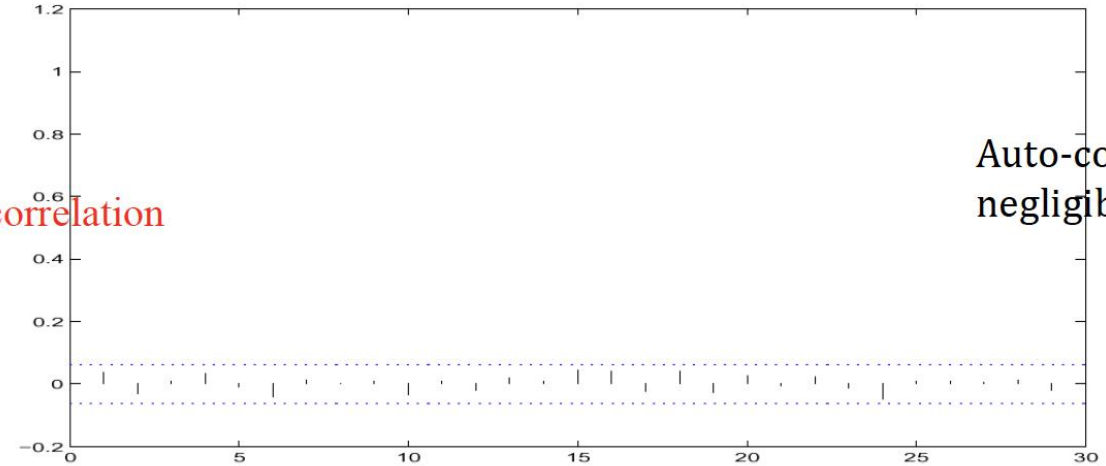
uniform qq plot



(a)

qq plot: straight line of 45 degrees if uniform -> TRUE

x_n appears to be uniform



autocorrelation

Test for independence -> TRUE

Auto-correlation is negligible

Period of RNG

This RNG appears to produce an iid, uniform output

However, it is in fact periodic

Any sequence generated from a deterministic algorithm in a given computer must be periodic (because the number of states is finite)

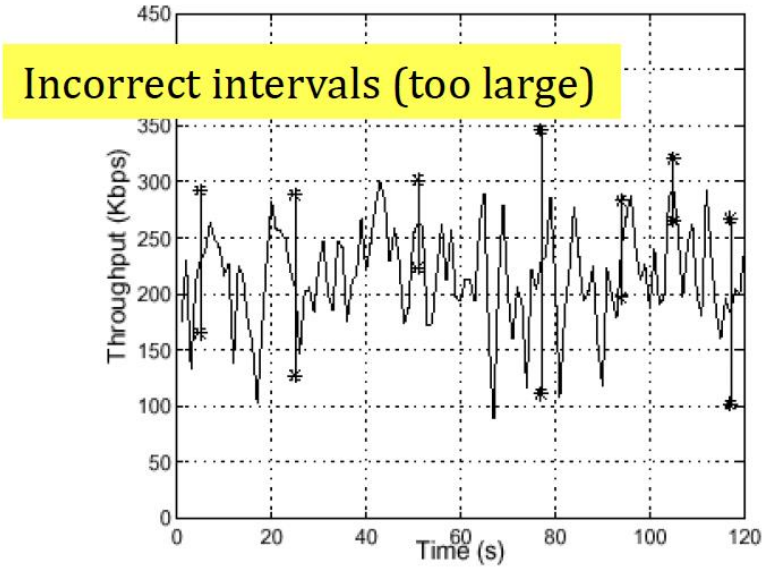
But we can require that the period should be much larger than maximum number of uses

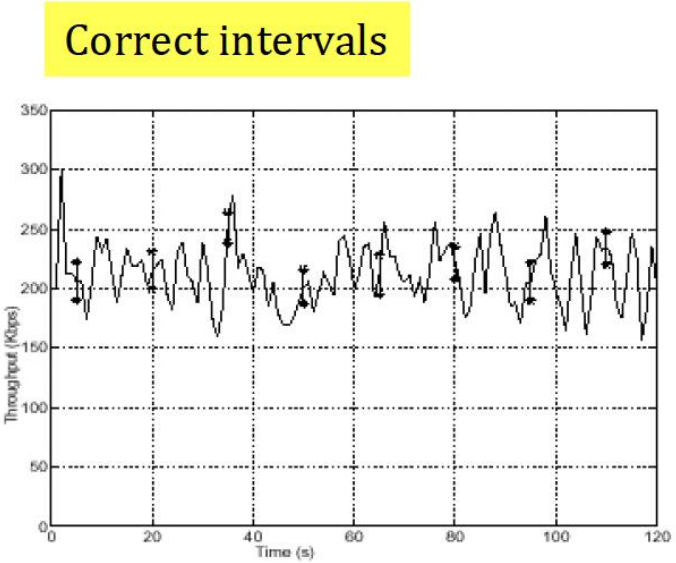
The ns2 simulator has period $\approx 2 \cdot 10^9$, which is too small

The "Mersenne twister" (matlab's default) has period $2^{19937} - 1 \approx 10^{6000}$

Example when the period is too small

Throughput of mobiles
Random numbers used for exponential backoff for WiFi (in collisions) and mobility models





(a) Linear Congruence with $a = 16'807$ and $m = 2^{31} - 1$ (b) L'Ecuyer"'s generator[LecuyerSimConf-01]

Example when the period is too small

(x_n, x'_n) for two parallel streams with the ns2 RNG

Example 6.10 Parallel Streams with Incorrect Seeds

Assume that we need to generate two parallel streams of random numbers. This is very frequent in discrete event simulations; we may want to have one stream for the arrival process, and a second one for the service process. Assume that we use the linear congruential generator of Example 6.9, and generate two streams x_n and x'_n with seeds $s = 1$ and $s' = 2$. Figure 6.7 shows the results: the two streams are strongly correlated. In contrast, taking s' = the last value x_N of the first stream does not give rise to this problem.

As mentioned above, more modern generators do not have this problem either.

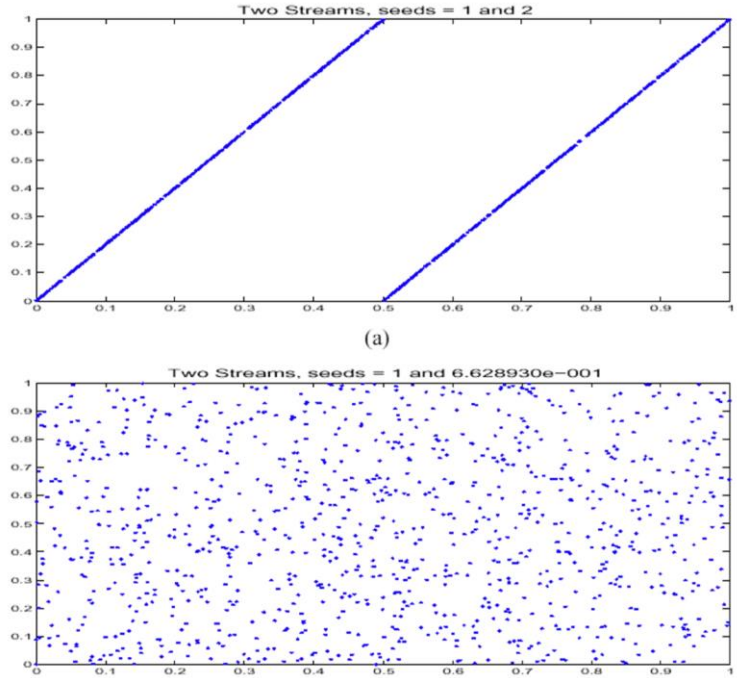


Figure 6.7 x_n versus x'_n for two streams generated with the linear congruential in Example 6.9. (a) Seed values are 1 and 2, (b) seed values are (1, last value of first stream).

Using RNGs

The RNG must have a period several orders of magnitude larger than the maximum number of times you will call it in the simulation.

The RNG uses a seed: for independent replications you need independent seeds; sequential runs usually are sufficient.

If you use parallel runs, you must find a way to obtain truly independent seeds:

- Eg based on computer clock
- Based on inputs by the user
- Using https://www.random.org/ generates true random numbers based on atmospheric noise
- Or by using an RNG that supports parallel streams, e.g., RandStream.create in Matlab

Sampling from A distribution

Problem is:

Given a distribution *F()*, and an RNG, produce some sample *X* that is drawn from this distribution

A common task in simulation

Matlab does it for us most of the time, but not always

Two generic methods:

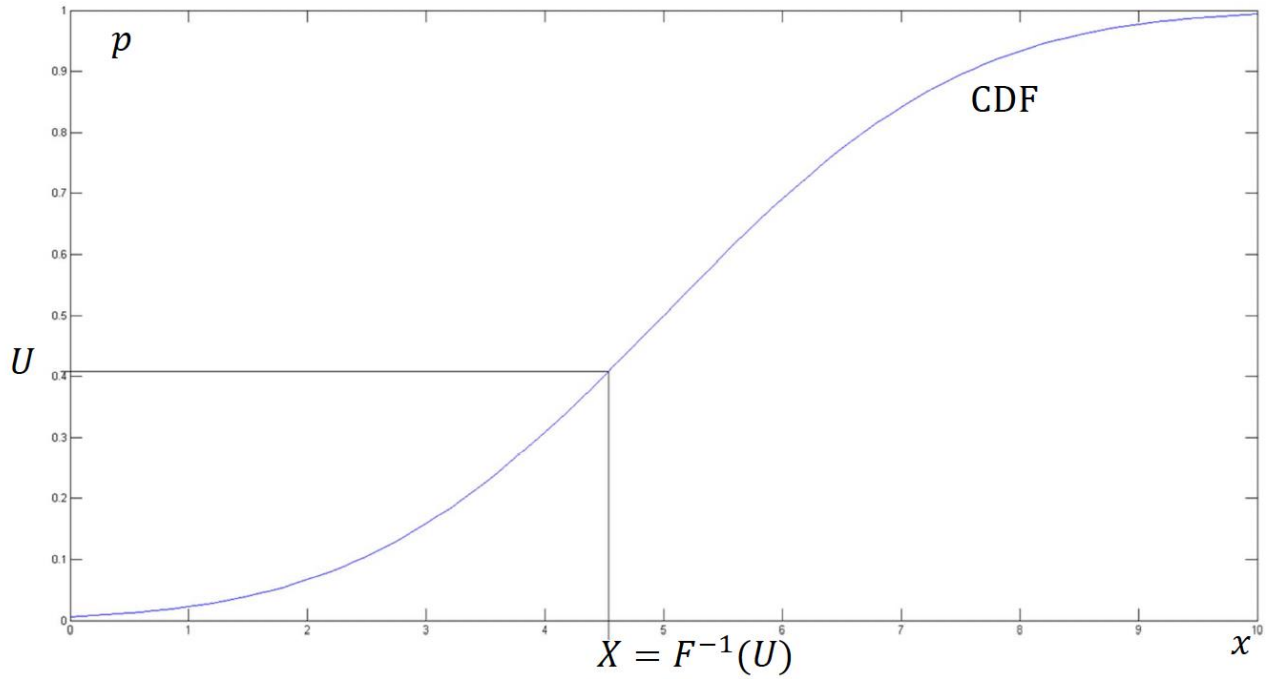
CDF inversion

Rejection sampling

CDF inversion for distribution with CDF F()

Applies to *real or integer* valued random variable

For a continuous distribution: draw *U~U(0,1)* using RNG; then $X = \mathrm{F}^{-1}(U)$ is drawn from the distribution with CDF *F()*



CDF inversion for distribution with CDF F() - Proof

U is uniformly distributed: $P(U \leq u) = u$ for $u \in [0,1]$

We define X by $X = \mathrm{F}^{-1}(U)$, i.e. $U = F(X)$
Let us compute the CDF of X

$(X \leq x) \Leftrightarrow \big(F(X) \leq F(x)\big)$ because F is monotonic increasing

Therefore

$$P(X \leq x) = P\big(F(X) \leq F(x)\big) = F(x)$$

CDF inversion for distribution with CDF F() - Example

EXAMPLE 7.10: EXPONENTIAL RANDOM VARIABLE. The CDF of the *exponential distribution* with parameter λ is $F(x) = 1 - e^{-\lambda x}$. The pseudo-inverse is obtained by solving the equation

$$1 - e^{-\lambda x} = p$$

where x is the unknown. The solution is $x = -\frac{\ln(1-p)}{\lambda}$. Thus a sample X of the exponential distribution is obtained by letting $X = -\frac{\ln(1-U)}{\lambda}$, or, since U and $1 - U$ have the same distribution:

$$X = -\frac{\ln(U)}{\lambda} \tag{7.9}$$

where U is the output of the random number generator.

Sampling an integer valued RV

We want to draw a sample of the random integer $\mathrm{N} \geq 0$ such that

$$p_k = P(N = k)$$

Let F_n=$\sum_{\mathrm{i}=0}^{\mathrm{n}} \mathrm{p_i}$ for $\mathrm{n} = 0,1,2,\ldots$ and $\mathrm{F}_{-1} = 0$.

1. Draw *U* using the RNG
2. Output *n* such that $\mathrm{F}_{\mathrm{n}-1} \leq \mathrm{U} \leq \mathrm{F}_{\mathrm{n}}$

Sampling an integer valued RV – Example

Ex1. Assume a random integer N taking values between 0 and 9 uniformly.
Then, $\mathrm{F}_{-1} = 0, \mathrm{F}_0 = 0.1, \mathrm{F}_1 = 0.2, \ldots, \mathrm{F}_9 = 1$

- Draw U using the RNG
- Assume U=0.22 then n=2
- Assume U=0.78 then n=7

Ex2. Poisson distribution with CDF $F_n = e^{-\lambda} \sum_{i=0}^{n} \frac{\lambda^i}{i!}$

✓ Draw U using the RNG

✓ Return n such that $e^{-\lambda} \sum_{i=0}^{n-1} \frac{\lambda^i}{i!} \leq U < e^{-\lambda} \sum_{i=0}^{n} \frac{\lambda^i}{i!}$

No closed form here

Sampling an integer valued RV - Example

EXAMPLE 7.11: GEOMETRIC RANDOM VARIABLE. Here X takes integer values $0, 1, 2, \ldots$. The *geometric distribution* with parameter θ satisfies $\mathbb{P}(X = k) = \theta(1-\theta)^k$, thus for $n \in \mathbb{N}$:

$$F(n) = \sum_{k=0}^{n} \theta(1-\theta)^k = 1 - (1-\theta)^{n+1}$$

$$F^{-1}(p) = n \Leftrightarrow n \leq \frac{\ln(1-p)}{\ln(1-\theta)} < n+1$$

hence

$$F^{-1}(p) = \left\lfloor \frac{\ln(1-p)}{\ln(1-\theta)} \right\rfloor$$

and, since U and $1-U$ have the same distribution, a sample X of the geometric distribution is

$$X = \left\lfloor \frac{\ln(U)}{\ln(1-\theta)} \right\rfloor \tag{7.11}$$

Question 3

What does this function compute ?

myfun($\boldsymbol{p}$) = if rand() $< p$ 0 else 1

A. The flip of a coin where 1 is obtained with proba p and 0 with proba $1-p$
B. The flip of a coin where 1 is obtained with proba $1-p$ and 0 with proba p
C. A sample of a geometric random variable with mean $\frac{1}{p}$
D. A sample of a geometric random variable with mean $\frac{1}{1-p}$
E. I don't know

Question 3 - Solution

What does this function compute ?

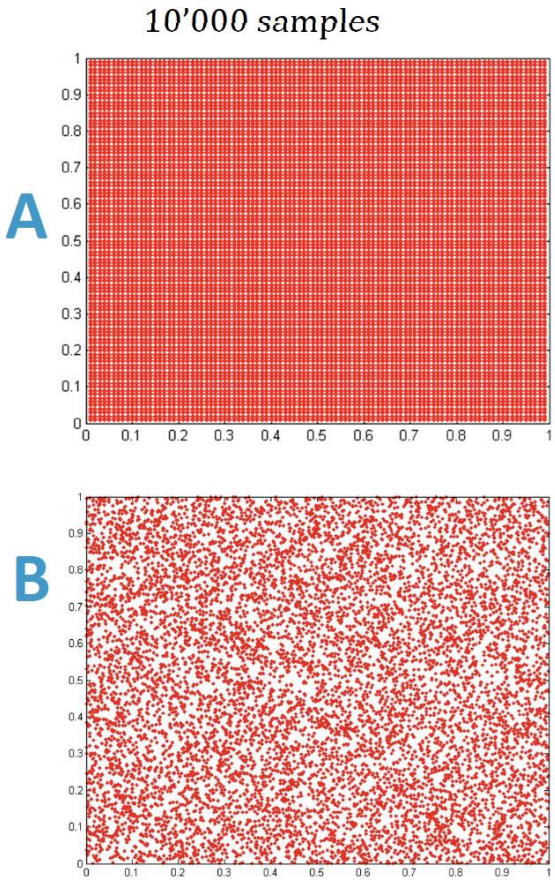
myfun($\boldsymbol{p}$) = if rand() $< p$ 0 else 1

A. The flip of a coin where 1 is obtained with proba p and 0 with proba $1-p$
B. The flip of a coin where 1 is obtained with proba $1-p$ and 0 with proba p
C. A sample of a geometric random variable with mean $\frac{1}{p}$
D. A sample of a geometric random variable with mean $\frac{1}{1-p}$
E. I don't know

Question 4: Sampling in two dimensions

Which one is a uniform random point?

A. A
B. B
C. Both
D. None
E. I don't know



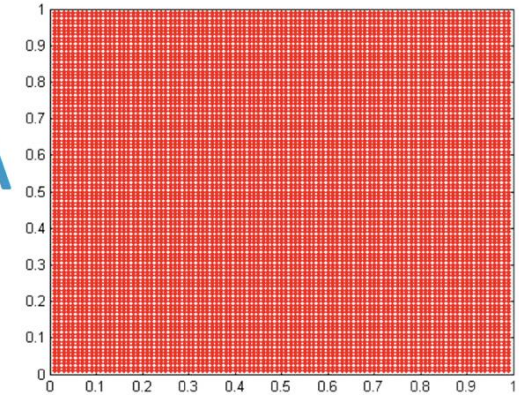
Question 4: Solution

Which one is a uniform random point?

A. A
B. B
C. Both
D. None
E. I don't know

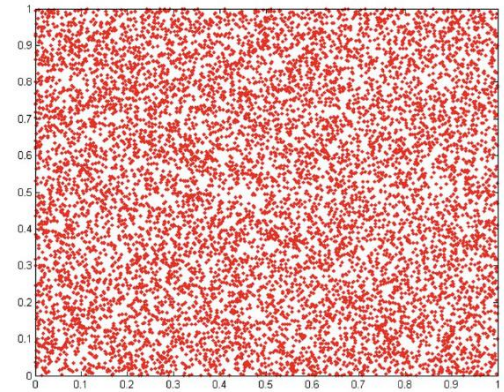
10'000 samples

A



Looks regular. Lines with equal spacing.

B



Sampled from uniform distribution.

How do you sample a uniform point in a rectangle ?

Let $[x_{min}, x_{max}]x[y_{min}, y_{max}]$ be the renctangle

M=(x,y) where X and Y are indepdendent because the distribution of Y given X=x is uniform in $[y_{min}, y_{max}]$ and therefore does not depend on x.

Do n times

Draw U,V from the RNG

X=(1-U) $x_{min} + U\, x_{max}$

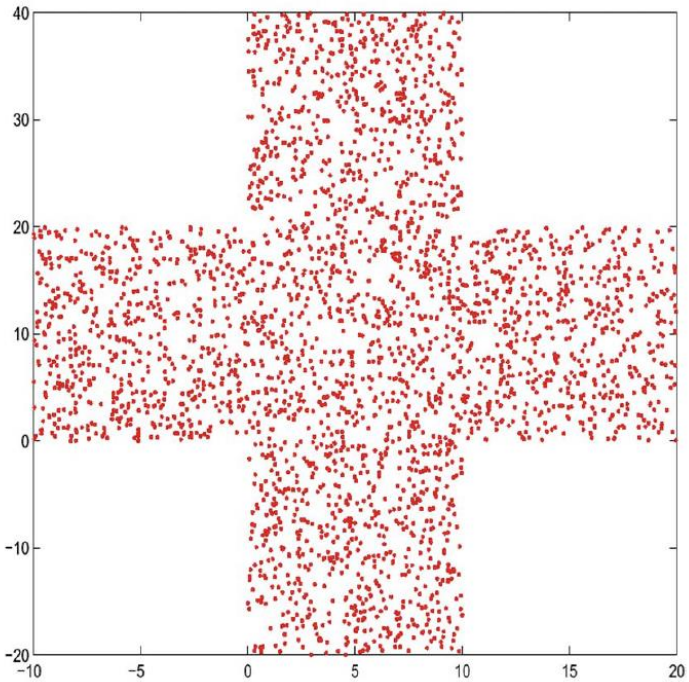
Y=(1-V) $y_{min} + V\, y_{max}$

end

Question 5

We sample a point $\boldsymbol{M} = (\boldsymbol{X}, \boldsymbol{Y})$uniformly in some arbitrary area. Are $\boldsymbol{X}$ and $\boldsymbol{Y}$ independent ?

A. Yes
B. It depends on the area
C. No
D. I don't know



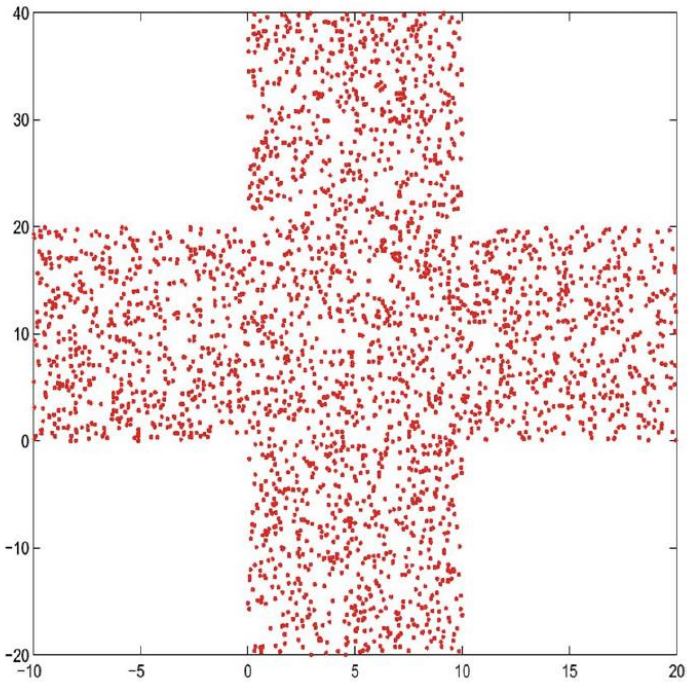
Question 5 - Solution

We sample a point $\boldsymbol{M} = (\boldsymbol{X}, \boldsymbol{Y})$uniformly in some arbitrary area. Are $\boldsymbol{X}$ and $\boldsymbol{Y}$ independent ?

A. Yes
B. It depends on the area
C. No
D. I don't know

For rectangle that we have seen before they are.

Here, not: for x=-5 y~U(0,20) and for x=5 y~U(-20,40)



Question 6

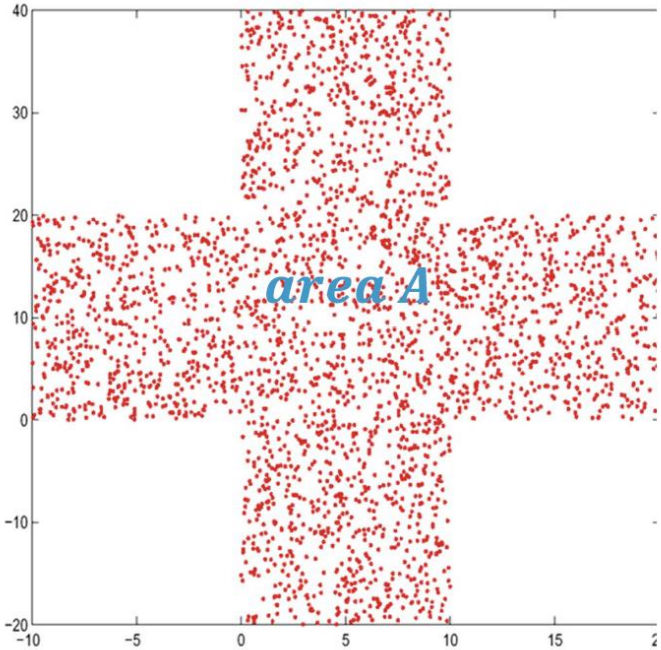
What is the PDF of the uniform distribution in A ?

A. $f_{X,Y}(x,y) = \begin{cases} 1 \text{ if } (x,y) \in A \\ 0 \text{ otherwise} \end{cases}$

B. $f_{X,Y}(x,y) = \begin{cases} xy \text{ if } (x,y) \in A \\ 0 \text{ otherwise} \end{cases}$

C. Something else

D. I don't know



Question 6 - Solution

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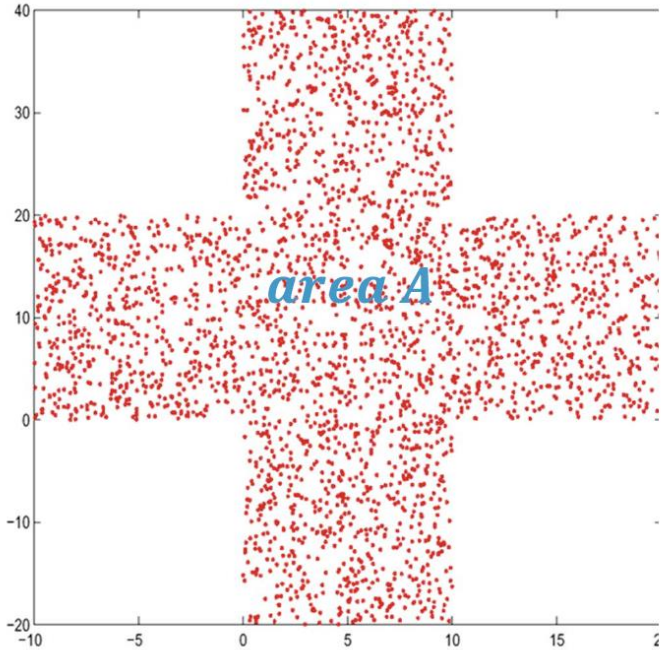
C. Something else

D. I don't know



The pdf is constant in A and zero outside A, however the constant should be such that the total probability is 1 :

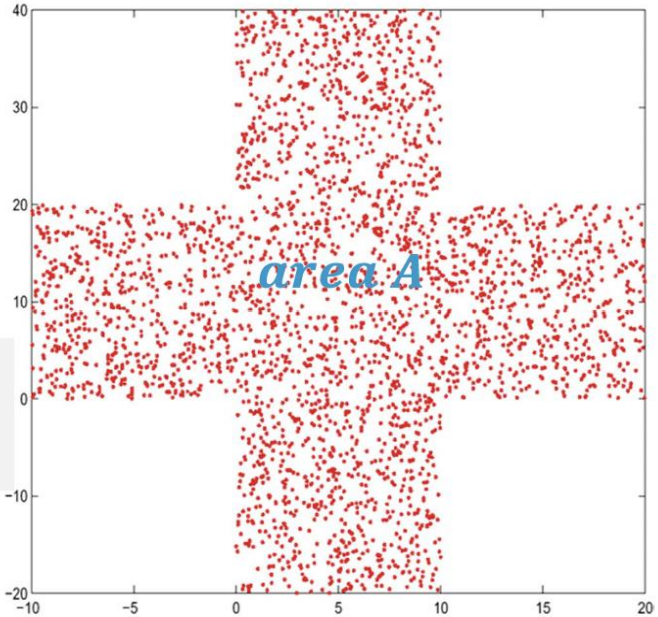
$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\text{area}(A)} \text{ if } (x,y) \in A \\ 0 \text{ otherwise} \end{cases}$$

Question 7

How can we sample a point uniformly distributed in this area $\boldsymbol{A}$?

A draw $X \sim U(-10,20)$
if $X \in [0;10]$ draw $Y \sim U(-20;40]$
else draw $Y \sim U(0;20]$

B repeat
draw $X \sim U(-10,20), Y \sim U(-20;40]$
until $(X,Y) \in A$



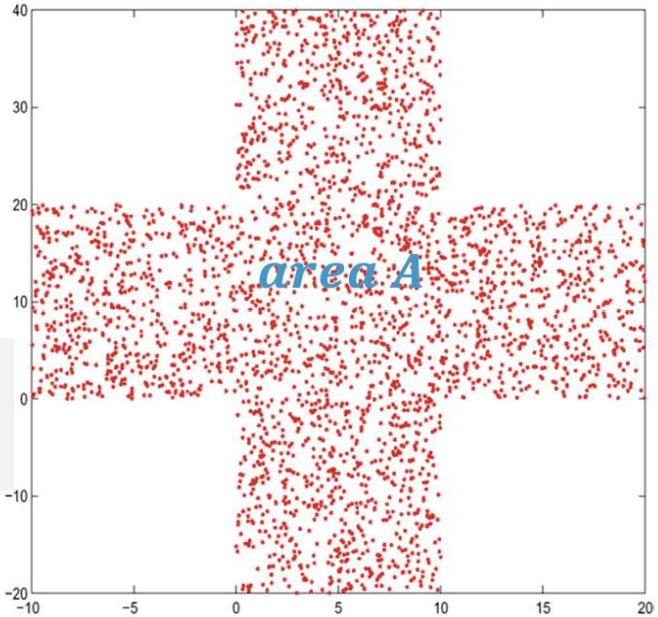
A. A
B. B
C. Both
D. None
E. I don't know

Question 7 - Solution

How can we sample a point uniformly distributed in this area $\boldsymbol{A}$?

A
draw $X \sim U(-10,20)$
if $X \in [0;10]$ draw $Y \sim U(-20;40]$
else draw $Y \sim U(0;20]$

B
repeat
draw $X \sim U(-10,20), Y \sim U(-20;40]$
until $(X,Y) \in A$



A. A
B. B
C. Both
D. None
E. I don't know

A is wrong because X is not uniform in [-10,20] but has higher chance of falling in [0,10]

B is correct and a method called rejection sampling: Gives the conidtional probability of (X,Y) given that $(X,Y) \in A$

Rejection Sampling for Conditional Distribution

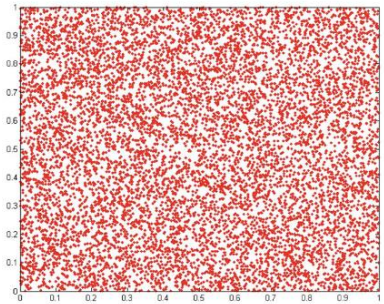
A uniformly distributed point in an area A has pdf equal to

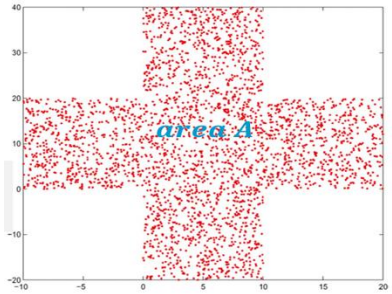
$$f_{xy}(x, y) = 1_{(x,y)\in A} \cdot \frac{1}{\text{area}(A)}$$

It can be sampled via rejection sampling;

1. Sample a point (X,Y) uniform in a bounding box around A
2. Reject (X,Y) if not in A
3. Until (X,Y) in A
4. Return (X,Y)

Note: The distributions in the two areas below differ!





Rejection Sampling for Conditional Distribution

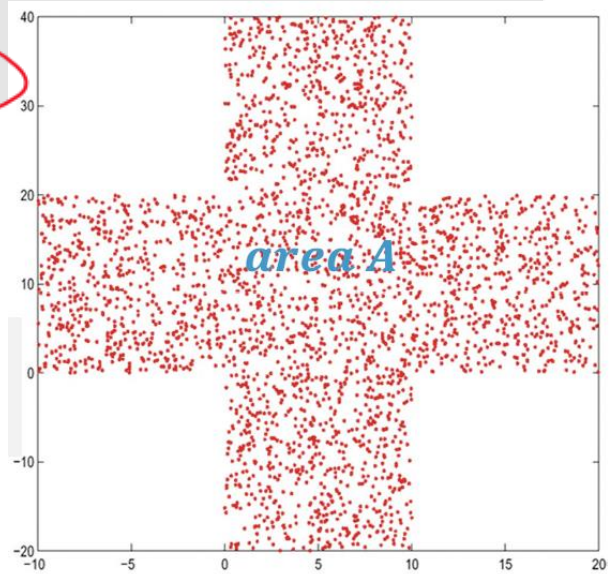
THEOREM 6.2 (Rejection Sampling for a Conditional Distribution). *Let X be a random variable in some space S such that the distribution of X is the conditional distribution of $\tilde{X}$ given that $\tilde{Y} \in \mathcal{A}$, where $(\tilde{X}, \tilde{Y})$ is a random variable in $S \times S'$ and $\mathcal{A}$ is a measurable subset of S.*
A sample of X is obtained by the following algorithm:

do
  draw a sample of $(\tilde{X}, \tilde{Y})$
until $\tilde{Y} \in \mathcal{A}$
return$(\tilde{X})$

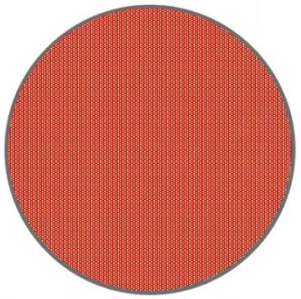
The expected number of iterations of the algorithm is $\frac{1}{\mathbb{P}(\tilde{Y} \in \mathcal{A})}$.

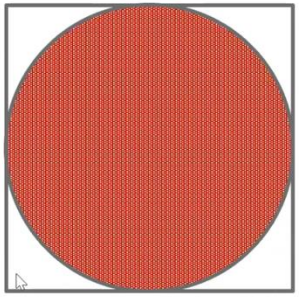
Notation in this theorem	Previous example
$\tilde{X}$	$(X, Y) \sim$ Unif(rectangle)
$\tilde{Y}$	$(X, Y) \sim$ Unif(rectangle)
X	$(X, Y) \sim$ Unif(A)

Here $\tilde{X}$ is the same as $\tilde{Y}$



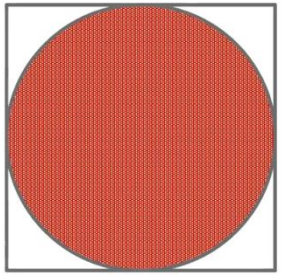
Examples



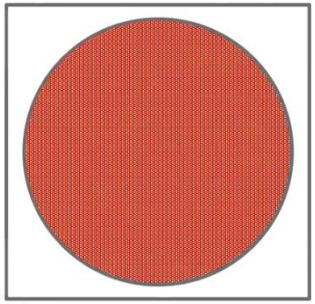


Use rejection sampling with a bounding box around the disk

sample a point (X, Y) uniform in the bounding box
reject (X, Y) if $X^2 + Y^2 > 1$
until $(X, Y) \in A$
return (X, Y)



Case A: square with side 2



Case B: square with side 2.2

How to choose?

Number of iterations: $\frac{1}{P(X^2+Y^2 \leq 1)}$

Case A: $\frac{4}{\pi}$ < Case B: $\frac{4.84}{\pi}$

Monte Carlo Simulation

A simple method to compute integrals and probabilities:

Idea:
Interpret the integral or probability β as $\beta = \mathrm{E}(\phi(\mathrm{X}))$
Simulate as many independent samples of X (using e.g., one of the previous methods) as you want and estimate the expectation by the mean

Method:
1. Generate *R* replicates X^r, *r=1,2,…R*
2. The Monte Carlo estimate of β *is*

$$\hat{\beta} = \frac{1}{R}\sum_{1}^{\mathrm{R}} \phi(X^r)$$

3. Compute a confidence interval for the mean because β is the mean of the distribution of $\phi(\mathrm{X})$

Monte Carlo Example (1)

Example: Compute $\beta =$

$$\int_0^1 \int_0^1 \int_0^1 \int_0^1 \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\, dx_1 dx_2 dy_1 dy_2$$

i.e. the average distance between two points uniformly distributed in the unit square $[0\,;1]^2$. Say which one is a correct implementation of the Monte-Carlo method:

A

1. Draw R points M^r, N^r random uniform in $[0,1]^2$
2. $\hat{\beta} = \text{mean}(d(M^r, N^r), r = 1{:}R)$
3. $\hat{\sigma} = \text{std}(d(M^r, N^r), r = 1{:}R)$
4. $CI = \hat{\beta} \pm 1.96\hat{\sigma}$

B

$b = 0, s = 0$
do R times
sample (X_1, Y_1, X_2, Y_2) iid $\sim$ Unif (0,1)
$b = b + \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2}$
$s = s + (X_1 - X_2)^2 + (Y_1 - Y_2)^2$
end
$\hat{\beta} = b/R$
$\hat{\sigma} = \sqrt{\frac{s}{R} - \hat{\beta}^2}$
$CI = \hat{\beta} \pm 1.96 \frac{\hat{\sigma}}{\sqrt{R}}$

A. A
B. B
C. Both
D. None
E. I don't know

Monte Carlo Example (2)

Example: Compute $\beta =$

$$\int_0^1 \int_0^1 \int_0^1 \int_0^1 \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\, dx_1 dx_2 dy_1 dy_2$$

i.e. the average distance between two points uniformly distributed in the unit square $[0\,;1]^2$. Say which one is a correct implementation of the Monte-Carlo method:

A

1. Draw R points M^r, N^r random uniform in $[0,1]^2$
2. $\hat{\beta} = \text{mean}(d(M^r, N^r), r = 1{:}R)$
3. $\hat{\sigma} = \text{std}(d(M^r, N^r), r = 1{:}R)$
4. $CI = \hat{\beta} \pm 1.96\hat{\sigma}$

B

$b = 0, s = 0$
do R times
sample (X_1, Y_1, X_2, Y_2) iid $\sim$ Unif $(0,1)$
$b = b + \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2}$
$s = s + (X_1 - X_2)^2 + (Y_1 - Y_2)^2$
end
$\hat{\beta} = b/R$
$\hat{\sigma} = \sqrt{\frac{s}{R} - \hat{\beta}^2}$
$CI = \hat{\beta} \pm 1.96 \frac{\hat{\sigma}}{\sqrt{R}}$

A. A
B. B
C. Both
D. None
E. I don't know

Because of the CI formula. In A the formula is a prediction interval. We will see the difference in later lectures. Otherwise both are correct. The circled formula is the 95% conf. interval for the mean.

Monte Carlo Example (3)

Compute $\beta =$

$$\int_0^1 \int_0^1 \int_0^1 \int_0^1 \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}\, dx_1 dx_2 dy_1 dy_2$$

i.e. the average distance between two points uniformly distributed in the unit square $[0\,;1]^2$.

We obtain the following Monte Carlo estimates

R	$\hat{\beta}$	95% confidence interval	
1'000	0.5254	0.5102	0.5405
10'000	0.5227	0.5178	0.5276
100'000	0.5206	0.5190	0.5221
1'000'000	0.5216	0.5211	0.5221

(The answer is known and is equal to 0.5214 [Ghosh B. 1943, On the distribution of random distances in a rectangle])

Confidence interval for the mean - Reminder

Theorem 2.2

Let $X_1, \ldots, X_n$ be n iid random variables, the common distribution which is assumed to have a well-defined mean μ and a variance σ^2. Let $\hat{\mu}_n$ and s_n^2 be

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{2.19}$$

$$s_n^2 = \frac{1}{n}\sum_{i=1}^{n} (X_i - \hat{\mu}_n)^2 \tag{2.20}$$

The distribution of $\sqrt{n}\frac{\hat{\mu}_n - \mu}{s_n}$ converges to the normal distribution $N_{0,1}$ when $n \to +\infty$. An approximate confidence interval for the mean at level γ is

$$\hat{\mu}_n \pm \eta \frac{s_n}{\sqrt{n}} \tag{2.21}$$

where η is the $\frac{1+\gamma}{2}$ quantile of the normal distribution $N_{0,1}$, i.e. $N_{0,1}(\eta) = \frac{1+\gamma}{2}$. For example, $\eta = 1.96$ for $\gamma = 0.95$ and $\eta = 2.58$ for $\gamma = 0.99$.

Conclusions

Simulating well requires knowing the concepts of

Transience
Confidence intervals
Sampling methods (Rejection sampling, CDF inversion)

Monte Carlo simulation is an efficient way to compute by simulation quantities that can be expressed as integrals